# PK-PD Practice 3

Module 7b - PK-PD Relationship · Daily practice sheet · No date

Name _____ Date _____

A 75-kg patient is to receive a single 20-mg/kg IV bolus dose of an antibiotic. The antibiotic follows a one-compartment open model, has an apparent volume of distribution that is approximately 0.3 L/kg, a first-order elimination half-life of approximately 3.6 hours, and is characterized by a minimum effective concentration of 7.2 mcg/mL.

**a** What is the expected duration of action of this antibiotic in this patient?

**b** What is the expected duration of action of this antibiotic in this patient if the dose were doubled?